Heart Disease Prediction using Machine Learning

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Abstract

Heart disease is one of the leading causes of death worldwide and has become a major public health concern. The increasing number of patients suffering from heart-related problems is mainly due to unhealthy lifestyle habits, lack of physical activity, poor diet, and stress. Traditional methods of diagnosing heart disease require several medical tests and expert analysis, which can be time-consuming and expensive. Moreover, human-based diagnosis may sometimes lead to errors due to complexity of medical data. With the growth of digital healthcare systems, large amounts of patient data are now available, which can be efficiently analyzed using machine learning techniques.

Keywords

Heart Disease Prediction, Supervised Learning, Machine Learning, Healthcare Analytics, Random Forest

Introduction

Heart disease is one of the most serious health problems affecting millions of people worldwide. According to global health organizations, cardiovascular diseases account for a large percentage of deaths every year. The early detection and prediction of heart disease are critical for improving patient survival rates and reducing healthcare costs. Traditional methods of diagnosing heart disease rely heavily on medical tests, clinical expertise, and manual evaluation of patient data. These processes can be time-consuming and sometimes prone to human errors. With the rapid growth of digital healthcare systems, large volumes of medical data are now available, making it possible to apply machine learning techniques for disease prediction.

Literature Review

Many researchers have explored the application of machine learning techniques for predicting heart disease, and the field has evolved significantly over time. Initially, traditional statistical methods were used for medical diagnosis, relying heavily on linear relationships and assumptions about data distribution. However, these approaches often struggled with complex and non-linear medical datasets, which limited their effectiveness in accurately predicting cardiovascular conditions [10].

The transition to machine learning introduced more flexible and powerful models such as Logistic Regression, Decision Trees, and Support Vector Machines. These algorithms were capable of handling non-linear relationships and large feature sets without strict assumptions about the data. Early studies demonstrated that classification algorithms could significantly improve prediction accuracy when applied to structured medical datasets like the UCI Heart Disease dataset [6][7]. In particular, Decision Tree models gained popularity due to their interpretability, allowing healthcare professionals to understand the reasoning behind predictions.

As research progressed, ensemble learning methods such as Random Forest and Gradient Boosting emerged as more robust alternatives. These methods combine multiple models to improve prediction performance and reduce overfitting. Several studies reported that Random Forest consistently outperformed individual classifiers in heart disease prediction tasks due to its ability to handle high-dimensional data and capture complex feature interactions [9][14]. This marked a shift from relying on single models to using ensemble-based approaches for better generalization.

With the advancement of deep learning, researchers began exploring neural networks for heart disease prediction. Deep learning models such as Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs) showed potential in capturing complex patterns within medical data. However, unlike image or speech data, structured healthcare datasets are often limited in size, which restricts the effectiveness of deep learning models. Studies have shown that for smaller datasets, traditional machine learning algorithms like Random Forest and Logistic Regression often perform comparably or even better than deep learning models due to lower computational requirements and better interpretability [25][26].

Another critical issue highlighted in the literature is the lack of standardization in datasets and evaluation metrics. Many studies use different datasets, preprocessing methods, and performance metrics, making it difficult to directly compare results across different research works [15][17]. Additionally, there is a possibility of bias in reported results, as studies with higher accuracy are more likely to be published.

Dataset Description

This research uses the **UCI Heart Disease Dataset**, which is widely used for machine learning research in healthcare. The dataset contains medical information of patients along with attributes related to heart disease diagnosis.

The dataset includes several important medical features such as:

- Age
- Sex
- Chest Pain Type

- Resting Blood Pressure
- Serum Cholesterol
- Fasting Blood Sugar
- Resting Electrocardiographic Results
- Maximum Heart Rate Achieved
- Exercise Induced Angina
- ST Depression
- Target (presence or absence of heart disease)

These attributes help machine learning models analyse patterns and determine whether a patient is likely to have heart disease.

Objectives

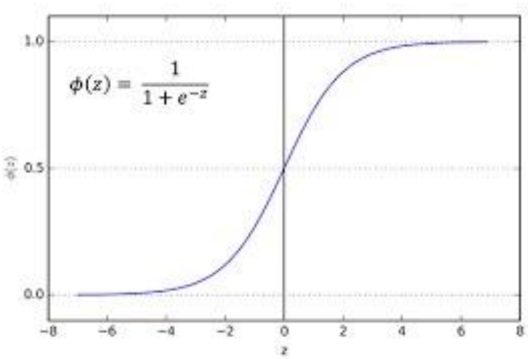
The main objective of this research is to develop an automated system for predicting heart disease using supervised machine learning models. The study aims to analyze patient medical data and identify important patterns that can help in early diagnosis. Another objective is to compare the performance of different supervised learning algorithms and determine the most accurate model for heart disease prediction.

Methods/Methodology

This research uses the UCI Heart Disease dataset, which contains medical attributes such as age, gender, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, and maximum heart rate. Data pre-processing techniques including handling missing values, normalization, and feature scaling were applied. The dataset was divided into training and testing sets. Four supervised learning algorithms were implemented using Python and the Scikit-learn library: The results show that ensemble learning models provide better prediction performance compared to individual classifiers.

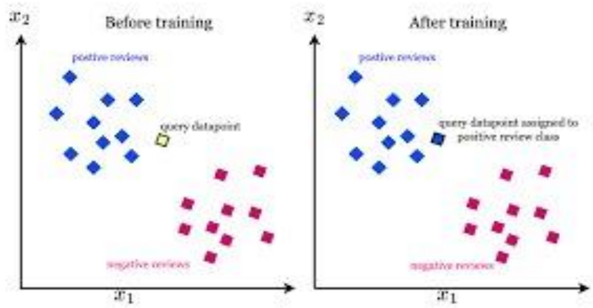
1. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems, where the output variable has two possible outcomes, such as the presence or absence of heart disease. Unlike linear regression, which predicts continuous values, logistic regression predicts the probability that a given input belongs to a particular class.



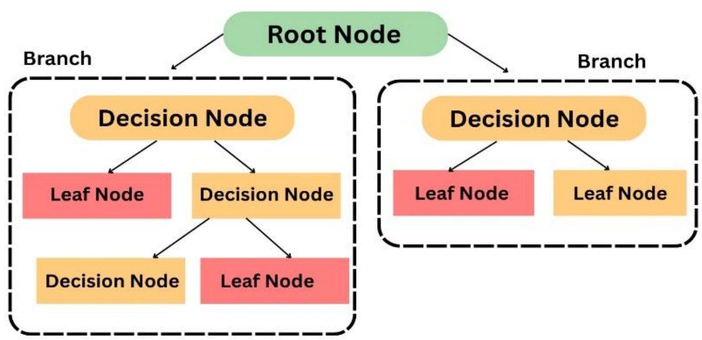
2. K-Nearest Neighbours (KNN)

K-Nearest Neighbours (KNN) is a supervised machine learning algorithm used for classification and regression tasks. It is a non-parametric and instance-based learning algorithm, meaning it does not make assumptions about the data distribution and stores all training data for prediction.



3. Decision Tree

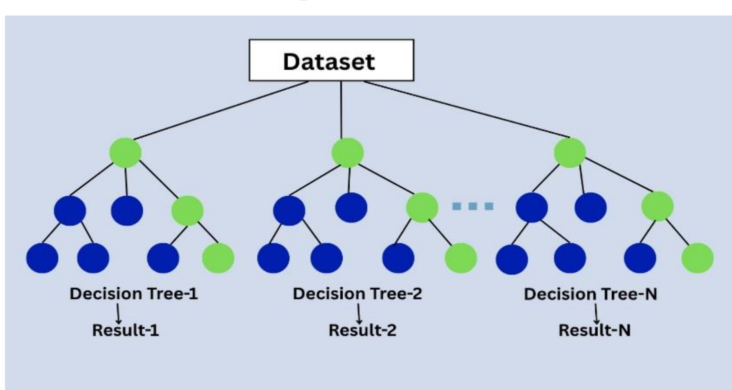
Decision Tree is a supervised machine learning algorithm used for classification and regression tasks. It is a tree-structured model where data is split into smaller subsets based on feature values, forming a structure similar to a flowchart. Each internal node represents a decision based on a feature, each branch represents an outcome, and each leaf node represents a final class label.



4. Random Forest

Random Forest is a supervised machine learning algorithm based on ensemble learning, where multiple decision trees are combined to improve prediction accuracy and reduce overfitting. Instead of relying on a single model, Random Forest builds a collection of decision trees and aggregates their predictions to produce a final result.

It is widely used for classification problems such as heart disease prediction because it can handle complex and non-linear relationships in data.



Results/Findings

The performance of all models was evaluated using accuracy as the main metric. Among the implemented models, Random Forest achieved the highest accuracy, followed by Logistic Regression and KNN.

Algorithm	**Accuracy**
Logistic Regression	82%
KNN	80%
Decision Tree	84%
Random Forest	89%

Conclusion/Future Work

This study concludes that supervised machine learning models can effectively predict heart disease with high accuracy. Among the evaluated models, Random Forest achieved the best performance due to its ability to reduce overfitting and handle complex data patterns. The proposed system can assist healthcare professionals in early diagnosis and improve decision-making processes.

In the future, this work can be extended by using larger and more diverse datasets, applying deep learning techniques, and integrating real-time patient data. Additionally, the system can be deployed as a web or mobile application for practical healthcare use.

References

[1] D. Dua and C. Graff, "UCI Machine Learning Repository," University of California, Irvine, 2019.

[2] R. Detrano et al., "International application of a new probability algorithm for the diagnosis of coronary artery disease," American Journal of Cardiology, vol. 64, no. 5, pp. 304–310, 1989.

[3] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[4] T. Hastie, R. Tibshirani, and J. Friedman, "The Elements of Statistical Learning," Springer, 2009.

[5] I. H. Witten, E. Frank, and M. A. Hall, "Data Mining: Practical Machine Learning Tools and Techniques," Morgan Kaufmann, 2011.

[6] S. Rajesh and T. Dhivya, "Analysis of heart disease prediction using machine learning algorithms," Int. J. Eng. & Tech., vol. 7, no. 2.8, pp. 684–687, 2018.

[7] K. G. Srinivas et al., "Applications of data mining techniques in healthcare," Int. J. Comput. Sci. Eng., vol. 2, no. 2, pp. 250–255, 2010.

[8] M. Akhil Jabbar et al., "Heart disease prediction system using associative classification," Int. Conf. Emerging Trends, 2017.

[9] S. Mohan et al., "Effective heart disease prediction using hybrid machine learning techniques," IEEE Access, vol. 7, pp. 81542–81554, 2019.

[10] J. Nahar et al., "Computational intelligence for heart disease diagnosis," Expert Systems with Applications, vol. 40, no. 1, pp. 96–104, 2013.

[11] H. Sharma and S. Sharma, "Machine learning based classification for heart disease prediction," Int. J. Comput. Appl., vol. 178, no. 7, 2019.

[12] U. K. Devi and S. S. Kumar, "Prediction of heart disease using machine learning," Int. J. Adv. Res. Comput. Sci., 2018.

[13] World Health Organization, "Cardiovascular diseases (CVDs)," 2023.

[14] A. Liaw and M. Wiener, "Classification and regression by randomForest," R News, vol. 2, no. 3, 2002.

[15] N. K. S. Kumar et al., "Heart disease prediction using ML: A survey," IJSCSEIT, 2020.

[16] P. K. Anooj, "Clinical decision support system for heart disease prediction using data mining," Int. J. Comput. Sci., 2012.

[17] R. Chitra and V. Seenivasagam, "Heart disease prediction system using supervised learning classifier," Bonfring Int. J., 2013.

[18] M. Gudadhe et al., "Decision support system for heart disease based on SVM and ANN," Int. Conf., 2010.

[19] K. Polat et al., "A hybrid approach to medical decision support system," Expert Systems, 2007.

[20] H. Kahramanli and N. Allahverdi, "Design of a hybrid system for heart disease diagnosis," Expert Systems with Applications, 2008.

[21] J. Soni et al., "Predictive data mining for medical diagnosis," Int. J. Comput. Appl., 2011.

[22] A. U. Haq et al., "A hybrid intelligent system for heart disease prediction," Mobile Information Systems, 2018.

[23] A. L. Bui et al., "Global burden of cardiovascular diseases," J. Am. College of Cardiology, 2011.

[24] S. Dilsizian and E. Siegel, "Artificial intelligence in medicine," J. Am. College of Radiology, 2014.

[25] R. Miotto et al., "Deep learning for healthcare," Briefings in Bioinformatics, 2018.

[26] Z. Obermeyer and E. J. Emanuel, "Predicting the future of big data in healthcare," Science, 2016.

[27] Y. LeCun et al., "Deep learning," Nature, vol. 521, pp. 436–444, 2015.

[28] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," KDD, 2016.

[29] J. Friedman, "Greedy function approximation: Gradient boosting machine," Annals of Statistics, 2001.

[30] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," JMLR, vol. 12, pp. 2825–2830, 2011.